

# Rule-Based Python Chatbot

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## Chatbot Code

```
# Rule-Based Python Chatbot

print("Chatbot: Hello! Before we start, what is your name?")
name = input("You: ").strip()

print(f"Chatbot: Nice to meet you, {name}! Type 'bye' or 'exit' anytime to end our chat.")

while True:
    message = input("You: ").lower().strip()

    if "bye" in message or "exit" in message:
        print(f"Chatbot: Goodbye, {name}! Have a great day.")
        break

    elif "hello" in message or "hi" in message:
        print(f"Chatbot: Hello there, {name}!")

    elif "how are you" in message:
        print("Chatbot: I'm doing great, thanks for asking! How about you?")

    elif "name" in message:
        print(f"Chatbot: You are {name}, and I'm just a simple chatbot!")

    elif "weather" in message:
        print("Chatbot: I can't check live weather, but I hope it's nice outside!")

    elif "help" in message:
        print("Chatbot: You can ask me about my name, the weather, or just say hello!")

    else:
        print(f"Chatbot: Sorry {name}, I don't understand that. Can you try again?")
```

## How the Code Meets Each Requirement

| # | Requirement               | Where in Code                    |
|---|---------------------------|----------------------------------|
| 1 | Ask the user's name first | input() before the main loop     |
| 2 | Use while True:           | Main conversation loop           |
| 3 | Use input()               | Collects each user message       |
| 4 | .lower().strip() cleaning | Applied to every message         |
| 5 | Two or more greetings     | "hello" and "hi"                 |
| 6 | Three or more topics      | how are you, name, weather, help |
| 7 | if / elif / else          | Full decision structure          |
| 8 | break on bye / exit       | First condition in the loop      |

| #         | Requirement            | Where in Code                                                         |
|-----------|------------------------|-----------------------------------------------------------------------|
| 9         | Uses the user's name   | Used in multiple responses                                            |
| 10        | else for unknown input | Final branch in the chain                                             |
| Challenge | Use of in operator     | Used throughout, so "hey hello" or "hello bot" still triggers a match |

## Suggested Test Inputs

- hello
- HELLO
- hi
- hey hello (challenge test using the in operator)
- how are you
- asdfgh (unknown message)
- bye or exit

*Target flow followed: INPUT → PROCESS → RESPOND → REPEAT → EXIT*